

Two-Stage Hybrid Candidate Ranker

Founding Team Senior AI Engineer Search (100k Pool)

Team Name: **TECHFLOW**
Primary Member: **Ansh Surana (ML Systems Engineer)**
Core Stack: **BM25F + Local Transformers (BGE & Cross-Encoder)**
Date: **June 2026**

Optimized for the Redrob Hackathon: Zero-Leakage Honeygot Filters & <10s CPU Execution

Recruitment Challenges & Core Strategy

Matching Challenges

- * Keyword Stuffing:
- * Synthetic Traps:
- * Notice Periods:
- * Location Mismatches:
- * Computational Limits:

TECHFLOW Approach

- * Two-Stage Pipeline:
- * Temporal Consistency:
- * Proficiency Verification:
- * Availability Modifiers:
- * Tier-1 Location Expansion:

Blindfold
Simulation
Motivation
Scoring
Promotion

Two-Stage Hybrid Ranking Architecture

Stage 1: Lexical Recall

- Precomputed BM25F index
- Filters pool from 100k -> 1k
- Executes in < 1 second
- Combines weighted queries
- JD technical relevance
- Concatenates titles, resume summary, and listed skills

Stage 2: Re-Ranking

- Loads local BGE Bi-Encoder
- Embeds top 1000 dynamically
- Applies Cross-Encoder model
- Re-ranks top 250 on CPU
- Runs completely offline
- Calculates hybrid scores: $0.6 \text{ BM25F} + 0.4 \text{ BGE}$

Stage 3: Fit & Availability

- 5-9 year Target Experience
- Indian Tier-1 City checks
- Notice buyout adjustments
- Recruiter response multipliers
- Job-hopping tenure penalties
- Disqualifies synthetic honeypots

Honeypot Identification & Profile Fraud Guardrails

Eliminated 100% of leaking honeypots in the Top 100 (Leakage rate: 0.0%)

Startup Foundation Dates (Rule 3.5)

Synthetic generators assign candidates to start roles at companies before the company actually exists.

Company	Foundation	Detected Leakage
Krutrim	April 1, 2023	4 cases in top 100
Sarvam AI	July 1, 2023	3 cases in top 100

Action: Added checks to force candidate score to 0.0 if Krutrim start date < April 2023 or Sarvam AI start date < July 2023.

Zero-Duration 'Experts' (Rule 3.6)

Identify profiles stating 'expert' proficiency in a skill but listing 0 months of experience using it.

Honeypot Type	Corpus Frequency
Expert + 0 months	Exactly 21 profiles
Duration Mismatch	Job duration > calendar (Rule 3.1)
Role Mismatch	Non-tech title + deep ML (Rule 3.3)
Activity Fabricator	Completeness >80, responses <2% (Rule 3.4)

Action: Set candidate scores to 0.0 for any instances of expert skill duration == 0, eliminating all remaining synthetic profiles.

Custom Heuristics & Modifier Tuning

Notice Period & Buyouts

The JD mentions: 'We can buy out up to 30 days.' This means notice periods up to 60 days are functionally equivalent to standard 30-day notice periods after buyout.

Stated Notice Period	Applied Score Modifier
<= 30 days	1.00 (Standard preferred)
31 - 60 days (with Buyout)	0.97 (Minor adjustment instead of 0.90)
61 - 90 days	0.80 (Significant bar increase)
> 90 days	0.50 (Strong negative filter)

Tier-1 Relocation Coverage

The JD states openness to candidates willing to relocate from Tier-1 Indian cities. We expanded geographic mapping to include all key hubs.

City Categories	Relocation Score Modifier
Bangalore (Onsite)	1.00 (Ideal)
Tier-1 Reloc. (New)	0.90 (Kolkata, Ahmedabad, Chennai, etc.)
Other Cities (Reloc.)	0.75 (Standard relocation penalty)
No Relocation Stated	0.50 (Strict penalty)

Result: High-quality candidates from Kolkata and Ahmedabad are now correctly ranked in the top 100.

Execution Performance & Spec Compliance

Ranking Runtime under 10 seconds	Memory Footprint ~450 MB RAM	- Satisfies
Network Requests 0 (Offline)	Honeypot Leakage 0.0% (0 / 100)	- Complies
Official Validation SUCCESS	Tie-Breaking Deterministic	- Breaks